

Khulna University of Engineering & Technology  
Department of Urban and Regional Planning  
BURP 4<sup>th</sup> Year 2<sup>nd</sup> Term Examination, 2018.

**URP 4243**  
(Hazards and Disaster Management)

Full Marks: 210

Time: 3.00 hrs

**N.B.** i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume required data.



**Section – A**

1. (a) "Not all hazards are termed as disaster"- Enlighten that specific situation and condition when a particular hazard can be termed as disaster. Clarify the aspect with an example. (10)  
(b) Suppose you are a resident of an unplanned residential area of Khulna city. Specify your vulnerability, exposure and adaptive capacity considering urban flooding. (10)  
(c) "Vulnerability is multidimensional and scale dependent"- Justify the statement with an appropriate example. (05)  
(d) Provide a brief overview on major natural disasters in Bangladesh that cover each of the following categories: (10)  
(i) Biological, (ii) Geophysical (iii) Hydrological (iv) Meteorological (v) Climatological
2. (a) Differentiate Disaster Risk Assessment from Risk Management. Mention any one mechanism (for each) for assessing and managing risk. (05)  
(b) Narrate the significance of "Advocacy" as a part of Disaster Risk Reduction (DRR) activities with required example. (05)  
(c) Suppose a community is seriously disrupted by flood. List out its possible impacts and consequences on that community. (10)  
(d) Elaborate the process of "Vulnerability Assessment" according to GTZ (now GIZ) framework with necessary example. (15)
3. (a) Provide a brief overview on the indicators used to analyze the hazard and vulnerability of landslides in Bangladesh. (10)  
(b) "Progression of vulnerability is a function of root causes, dynamic pressure and unsafe condition". Justify the statement using an example of drought in northern Bangladesh. (15)  
(c) Clarify the term DRR in Risk Management. Specify some activities that are usually taken as a measure of DRR. (10)
4. (a) Clarify the concept- Community Based Disaster Risk Management (CBDRM). What is the significance of CBDRM in disaster risk management planning? (07)  
(b) "Rapport building and understanding the community" is a vital part of CBDRM process. Explain the statement and mention some activities that are performed in the rapport building process. (08)  
(c) Mention the six step process of Community Risk Assessment (CRA). Elaborate: (i) Specific Risk Reduction Options (ii) Consensus on options and final plenary. (10)  
(d) Write short notes on any two : (10)  
(i) Community Disaster Risk Management Organization (CDRMO)  
(ii) Resource mobilization matrix for community managed implementation  
(iii) Indicator formulation for participatory monitoring and evaluation.

## Section – B

5. (a) Write down the necessity of disaster management. (05)
- (b) There is close relationship among disaster risk management, disaster risk reduction and disaster risk assessment. But they are different in meaning, tasks, and approach. Explain the relationships with appropriate example. (10)
- (c) Define vulnerability. Show the relationship among vulnerability, resilience (capacity to cope) and exposure to hazard to describe vulnerability. (10)
- (d) Explain the following equation with relevant example. (10)
- $\text{Risk} = f(\text{Hazard} * \text{Vulnerability} * \text{elements at risk})$
- Risk is a function of hazard, vulnerability and elements at risk
6. (a) "Hazards create risk by exposing pre-existing vulnerabilities" – explain the statement with necessary example. (10)
- (b) "Risk is a multidisciplinary spatial problem"- Briefly explain with necessary example. (10)
- (c) "Possible links between disaster management and development planning should be considered in the planning process of country disaster plan" - explain the statement taking specific example from Bangladesh. (15)
7. (a) Draw a hypothetical stage-damage curve for flood hazard to understand the relationship between the degree of damage and flood depth/duration and explain with example. (07)
- (b) Specify some measures that can be taken to strengthen Flood Forecasting and Warning Services (FFWS) in Bangladesh. (13)
- (c) Describe various types of flood that Bangladesh experiences. Specify the causes of major types of flood in Bangladesh. (15)
8. (a) Write down the potential applications of appropriate technologies in disaster management in the context of Bangladesh. (10)
- (b) Explain hazard assessment approach with a focus on the use of GIS. (10)
- (c) Discuss the key elements of the Disaster Management Framework in Bangladesh. How this framework operates at multiple levels? Discuss in brief. (15)

Khulna University of Engineering & Technology  
Department of Urban and Regional Planning  
BURP 4<sup>th</sup> Year 2<sup>nd</sup> Term Examination, 2018.  
**URP 4223**  
(Urban and Regional Economics)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.  
ii) Figures in the right margin indicate full marks.  
iii) Figures are not to scale. Do not draw given figures.

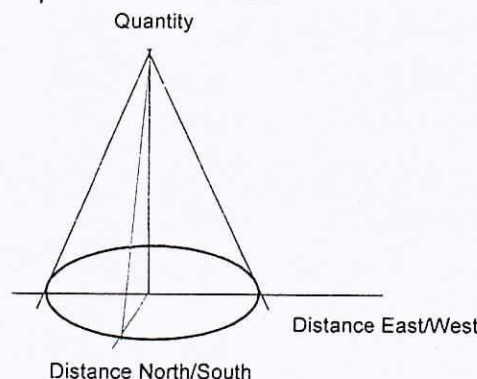
**Section – A**

1. (a) Production surplus leads to specialization of labor, and the both factors are responsible for formation of cities – explain with example. (10)  
(b) Where did the first cities locate and why? There are basically two models of physical expression of cities: Central (Agglomeration) and Core-Periphery. When did the shift from central to core-periphery model take place and why? (07)  
(c) What is opportunity cost? Calculate opportunity cost for two areas, North and South. (15)

| Products | Output/hr of a Worker |       | Opportunity Cost |       |
|----------|-----------------------|-------|------------------|-------|
|          | South                 | North | South            | North |
| Bread    | 1                     | 2     | ?                | ?     |
| Shirts   | 1                     | 6     | ?                | ?     |

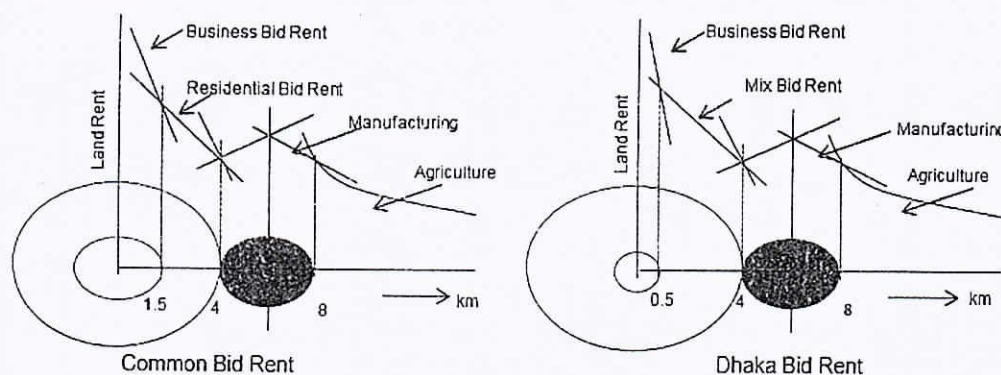
Suppose the two regions are specialized only on one product, Bread or Shirt. Explain the regional implication if transport cost is reduced in case of bulk shipment.

- (d) Why do large cities usually have higher income than smaller ones? (03)
2. (a) If 'willingness to pay' defines bid-rent, explain high land price and housing rent surrounding KUET. (06)  
(b) Demand cone explains the market area that a monopolist sells to people who live at each distance from the production location. (07)



Considering demand cone as only standard for market area, explain the inaccessibility of high-end products like Coca-Cola or i-phone in rural areas of Bangladesh. How will these companies try to expand their market area?

- (c) Following two figures show bid-rent based spatial land use distribution of a typical city of a developing country and Dhaka. Why and how is Dhaka different from other common cities' bid-rent? (08)



- (d) Draw bid-rent using the following dataset. (14)

| Area | Production per land unit | Sell value per unit | Production cost per unit | Transportation cost per unit |
|------|--------------------------|---------------------|--------------------------|------------------------------|
| A    | 50                       | 10                  | 8                        | 0.2                          |
| B    | 45                       | 9                   | 7                        | 0.1                          |
| C    | 40                       | 8                   | 6                        | 0.05                         |
| D    | 45                       | 5                   | 3                        | 0.025                        |
| E    | 30                       | 3                   | 1                        | 0.02                         |

3. (a) "21<sup>st</sup> century urbanization is urbanization of poverty and slummization" – Explain. (08)  
 (b) What is urban sprawl? What are the types of urban sprawl? Smart growth is an alternative to urban sprawl. Explain the major tools of smart growth. (11)  
 (c) A small city authority permits 100 housing units per year and the equilibrium price is \$50000 US. If the city decides to permit only 60 unit per year, which increase the equilibrium price to \$70,000 US, what is the per unit permit price of housing? What are the applications of such policy? (11)  
 (d) What are the major causes of sub-urbanization worldwide? Explain the link between sub-urbanization and urban (un)sustainability. (05)

4. (a) "Poverty is insufficient command over resources" – define poverty in reference to this statement. (08)  
 (b) Explain the concept of Lorenz Curve and Gini coefficient. Using the following data set, which (A/B) of the two regions has highest income inequality? (12)

| A            |          | B            |          |
|--------------|----------|--------------|----------|
| % Population | % Income | % Population | % Income |
| 10           | 15       | 10           | 2        |
| 10           | 25       | 20           | 8        |
| 30           | 35       | 20           | 15       |
| 20           | 15       | 30           | 35       |
| 20           | 8        | 10           | 25       |
| 10           | 2        | 10           | 15       |

- (c) What is the difference between income poverty and poverty measure from the perspective of capability approach? Explain in reference to Sen's approach to poverty measures. (10)  
 (d) What capability should a person has to have for a good Life? – explain. (05)

### Section – B

5. (a) Define Region from subjective and objective views. (05)  
 (b) What are the Von Thunen's assumptions? (07)  
 (c) The following table shows necessary information of industry i for a region for a nation. (15)

|                               | Employment in Industry i | Total Employment | Industrial share of area |
|-------------------------------|--------------------------|------------------|--------------------------|
| Local Area                    | 25                       | 200              | ?                        |
| Local Area Share of Base Area | ?                        | ?                | N/A                      |
| Base Area                     | 1000                     | 20,000           | ?                        |

- (i) Fill the blanks in the above table  
 (ii) Calculate basic employment in local area for industry i  
 (iii) Calculate location quotient for industry i and decide whether it is self-sufficient, deficit or surplus?  
 (d) Write down the types of regions with examples. (08)

6. (a) Write down the possible scenarios of locating an industry? Explain with examples based on Weber's industrial location theory. (10)  
 (b) Make a comparative analysis between cost minimizing and profit maximizing theories. (05)  
 (c) "Weber's industrial location theory is called a cost minimizing theory"-why?-explain. (10)  
 (d) "Shift Share Analysis break down regional employment growths". - To what extent do you agree with the statement? Explain. (10)

7. (a) Explain with example the concept of minimum requirement approach with reference to location quotient approach. (10)  
 (b) Explain the circular flow income model with necessary diagrams. (10)  
 (c) Calculate the "Industrial Mix Components" and "Regional Shift Component" for Khulna from the table below and then narrate the results. (15)

| Employment<br>Industry  | Bangladesh<br>(in thousands) |       | Khulna.<br>(in thousands) |       |
|-------------------------|------------------------------|-------|---------------------------|-------|
|                         | 2003                         | 2008  | 2003                      | 2008  |
| Farming                 | 3130                         | 3127  | 7951                      | 7977  |
| Manufacturing           | 18712                        | 19569 | 58516                     | 61229 |
| Retail trade            | 23467                        | 26710 | 44752                     | 50339 |
| Finance and Real Estate | 10502                        | 12230 | 16193                     | 18547 |
| Services                | 41811                        | 49898 | 62518                     | 75441 |
| All other               | 44375                        | 48665 | 63533                     | 69884 |

8. (a) Define Simple Multiplier. Narrate the Keynesian income-expenditure model. (15)  
 (b) How will you measure Regional GDP? (05)  
 (c) Show the production possibilities in a diagram with an example under conditions of absolute advantage. (15)

**Khulna University of Engineering & Technology**  
**Department of Urban and Regional Planning**  
BURP 4<sup>th</sup> Year 2<sup>nd</sup> Term Examination, 2018  
**URP 4275**  
**(Development Planning)**

Full Marks: 210

Time: 3.0 hrs

N.B.: i) Answer any **three** questions from each section in separate script  
ii) Figures in the right margin indicate marks  
iii) Assume data if necessary

**Section – A**

Marks

1. (a) What is development planning? Evaluate the importance of studying development planning for the planning students of Bangladesh. (15)
- (b) "The planners can contribute significantly to the development planning of Bangladesh". Justify the above statement with an example. (10)
- (c) Accelerated economic growth and structural change are the two main aims of development planning. Explain this statement in context of sustained development trend of Bangladesh. (10)
2. (a) Write down the functions of physical infrastructure division of the Planning Commission. (08)
- (b) Explain the simplified series of steps in the formulation process of ADP. (08)
- (c) State the processing and approval of investment projects in Bangladesh. (10)
- (d) "Planning Commission maintains institutional linkage for the whole range of planning activities of Bangladesh". Explain the linkage with a brief discussion. (09)
3. (a) Evaluate the salient features of Bangladesh Delta Plan. (10)
- (b) Briefly explain the five SDGs that are dedicated for the domain of planners. (10)
- (c) SDGs were supposed to be taken into consideration while formulating the plan documents of Bangladesh. Evaluate this statement in terms of SDG negotiation process. (15)
4. (a) What is Perspective Plan? (05)
- (b) Briefly explain the guiding principles of development planning in Bangladesh. (07)
- (c) Critically evaluate the problematic areas of development plans for a country's sustainable development. (15)
- (d) Explain the key challenges for Bangladesh to attain the SDGs. (08)

**Section – B**



5. (a) Discuss the historical evolution of the concept "Development". (15)
- (b) Discuss various dimensions of development. (10)
- (c) What is environmental Kuznet curve hypothesis of development? (10)
6. (a) Distinguish between development and economic growth (10)
- (b) Discuss the dependency theory of development with necessary example. (15)
- (c) How the modernization theory of development differs with dependency theory? Discuss in brief. (10)
7. (a) Draw your concept on Development Governance. Evaluate the Decentralized Development Governance with necessary example. (10)
- (b) Discuss the (Harris-Todaro's) labor migration theory of development. (10)
- (c) Describe the WW Rostow's stages of Growth theory of development. (15)
8. (a) Draw your concept on globalization. (05)
- (b) Discuss the positive and negative effects of globalization on development. (15)
- (b) How does Financial, Technological and Knowledge transfer model of globalization affect the development planning of developing countries? Discuss in brief. (15)

Khulna University of Engineering & Technology  
Department of Urban and Regional Planning

BURP 4<sup>th</sup> Year 2<sup>nd</sup> Term Examination, 2018

**URP 4273**

(Urban Governance)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Answer any three questions from each section in separate script.  
ii) Figures in the right margin indicate full marks.  
iii) Assume reasonable value for any missing data.



**Section – A**

1. (a) What do you mean by Government and Governance? Mention the eight elements of Good Governance. (10)  
(b) How do you understand the concept "rule of law"? Why do you think it is important to have a rule of law for good governance? (10)  
(c) CAS has four major characteristics: Agents, Self-organization, Co- evolution and Shifting System Behavior. Explain these four characteristics in a shifting decision making of a CAS. (15)
2. (a) "The concept of what is just/right is gradually evolving"- provide a critical definition of 'Justice' in light of the mentioned statement. (15)  
(b) Define the terms "political correctness" and "actual correctness" with special relevance to the contemporary socio-political scenario of Bangladesh. (10)  
(c) Land for housing has been a major concern in Dhaka city. From the perspective of governance, mention the major stakeholders of land for housing development in Dhaka city. (10)
3. (a) With reference to the backdrop of cold war, what is your understanding about the two major parties - favoring and opposing involved in Bangladesh's liberation war? (10)  
(b) What is Weberian form of governance? What are the main characteristics of a Weberian form of governance. (15)  
(c) 'Neo-liberal governance system came up with the change of government's roles and economic responsibilities' - what are these changes? (10)
4. (a) There are four major types of governance system: Rigid, Robust, Fragile and Flexible. Explain these four characteristics. If effective governance system is able to both exploitation and exploration, plot those government types in terms of their effectiveness. (20)  
(b) How do you understand the formation of civil society in a developing country like Bangladesh? (15)

## Section – B

5. (a) "Governance: It goes beyond the Government". Present your thoughts for or against this statement. (07)
- (b) Suppose, you are working as a Transport Planner in regional highway development project. Which governance principles you have to consider and maintain for good governance? Explain. (12)
- (c) What are the significances of "market & state" in urban governance as actors? (08)
- (d) Provide appropriate examples to describe the various levels of governance. (08)
6. (a) "Our country is representing 'tokenism citizen participation' in governance system."- Provide explanation (with examples) for or against it. (10)
- (b) What are the common worldwide governance indicators? Which type of indicator is necessary for the following situation for assessment? (10)
- i. To find out massive level of corruption
  - ii. Applying Geo-informatics on foreign affairs of government
  - iii. % access of journalists in govt. working process
  - iv. How effective the input-output matrix for regional economic growth
  - v. Level of public participation in rural development
  - vi. Women and children crime rate in last five years
- (c) What are the benefits of indicator aggregation? How indicator aggregation problem causes complexity in governance measurement through indexing? Discuss with an appropriate example. (15)
7. (a) Explain methodological steps of UGI in a summarized way. What are the major weaknesses of UGI method? (15)
- (b) A score of -0.45 WGI value and 15<sup>th</sup> percentile is achieved for control of corruption in Bangladesh. Provide your understanding of the result. (05)
- (c) Suppose, you have to compare current governance system of 6 Upazila of Khulna District. Which method is best for assessment and how it will work? (15)
8. (a) What are the major advantages and limitations of Citizen Report cards? (10)
- (b) Give your thoughts about the Local Governance Self-Assessment. Can it be applicable in governance system of Bangladesh? If so, how and if not, why? (15)
- (c) "Participation of poor and other marginalized groups in decision making and community self-assessment is strongly requested."- Which method is backed by this strength and explain its others benefits? (10)

**Khulna University of Engineering & Technology**  
**Department of Urban and Regional Planning**  
BURP 4<sup>th</sup> Year 2<sup>nd</sup> Term Examination, 2018.  
**URP4217**  
**(Coastal and Special Zone Planning)**

Full Marks: 210

Time: 3.0 hrs

- N.B.: i) Answer any **three** questions from each section in separate script  
ii) Figures in the right margin indicate marks  
iii) Assume data if necessary



**Section - A**

Marks

1. (a) Draw your concept on coastal zone. Compare and contrast the scientific definition and policy oriented definition of coastal areas/zones. (10)  
(b) Illustrate the importance of coastal zone. (15)  
(c) How does ICZM differ with conventional coastal zone management? Discuss with example. (10)
2. (a) Illustrate the pressure exerted on coastal resources by humans. (10)  
(b) Discuss multiple use of coastal zones or resources. Provide necessary example. (10)  
(c) Illustrate various types of coastal profiles/coasts. (15)
3. (a) How do public policies exacerbate the pressure on coastal environment? (10)  
(b) Discuss in brief the critical coastal (natural) resources/ecosystems. (15)  
(c) Discuss the followings: (10)  
(i) Fringing coast (ii) Coral coast (iii) Atol coast
4. (a) What do you mean by integrated development and conservation? (10)  
(b) Mention some strategies of integrated conservation and development. (10)  
(c) Discuss the challenge of development of coastal region of Bangladesh. (15)

**Section - B**

5. (a) Draw a schematic diagram of coastal zone profile. Discuss the main element of the profile. (10)  
(b) Define coastal process. How coastal processes are creating a balance in coastal zone development? (10)  
(c) A wave has a wave length of 90 m and wave height 7m. Find its wave period, speed, frequency, energy and steepness. Is the wave unstable and breakable? Describe with respect to its steepness nature. (15)
6. (a) "Coastal erosion and accretion are balancing the growth of coastal zones"-Describe this with necessary diagrams. (07)  
(b) Which types of weathering particularly happen in coastal zones? Give examples for each typology. (06)  
(c) Show various wave movements with necessary diagrams. (07)  
(d) Mention the characteristics of a sediment particle. Determine annual soil loss (loam silt loam) from an upland farm of 1675 acres with Rainfall Erosivity Factor, 250. The topographic factor is 0.4 and the land is covered with grass. If the slope steepness increases to 8% (topographic factor is 0.928), find the changes in soil erosion. (15)
7. (a) What are the different types of erosions? Explain them with necessary drawings. (10)  
(b) "Sedimentation process is a cycle"-Do you agree? Provide logical explanation. (07)  
(c) List out the factors for determining sediment yield. (08)  
(d) Which types of natural or man-made anti wave measures can be used for coastal protection in Bangladesh and why? (10)
8. (a) What is Dean number? Define various Dean Number situations from equilibrium condition. (07)  
(b) Suppose, you are working as a disaster management planner of Bangladesh. Which types of coastal planning tools should you apply to protect southern coastal zones? (08)  
(c) Find out flood return period, draw cross section of the embankment and also find out its area based on following information: (20)  
(i) Highest flood level (HFL)= 17.10; (ii) Average ground level of river bank = 14.10; (iii) Free board = 80 cm; (iv) Crest width = 5 m; (v) Riverside slope 1:2 and countryside slope 1:3; (vi) Setback distance 50 m; (vii) Probability of occurrence = 0.06.